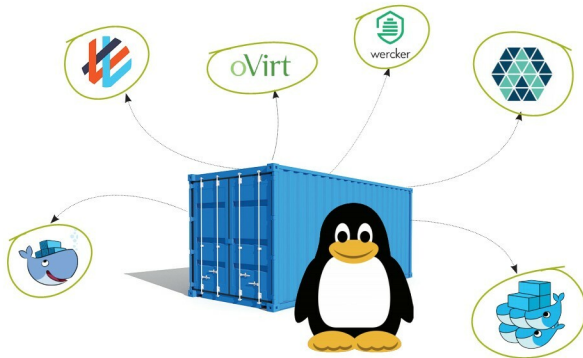


The Best Open Source Tools for Containerisation

Container management and automation tools are today a great necessity, since IT organisations need to manage and monitor distributed and cloud/native based applications. Let's take a look at some of the best open source tools that are available to us today for containerisation.



The Linux operating system, LXC, the building block that sparked the development of containerisation technologies, was added to the Linux kernel in 2008. LXC combined the use of kernel cgroups, which allow groups to be separated so that they cannot “see” each other, to implement lightweight process isolation.

Recently, Docker has evolved into a powerful way of simplifying the tooling required to create and manage containers. It basically uses LXC as its default execution driver. With Docker, containers have become accessible to novice developers and systems administrators, as it uses simple processes and standard interfaces.

Containerisation is regarded as a lightweight alternative to the virtualisation of the full machine, which involves encapsulating an application in a container with its own operating environment. This, in turn, provides many unique advantages of loading an application into a virtual machine, as applications can run on any suitable physical machine without any dependencies.

Containerisation has become popular through Docker, for which containers are designed and developed in such a manner that they are capable of running on everything ranging from physical machines and virtual machines, to

OpenStack cloud clusters, physical instances and all sorts of servers.

The following points highlight the unique advantages of containers.

- *Host system abstraction:* Containers are standardised systems, which means that they connect the host machine to anything outside of the container using standard interfaces. Container applications rely on host machine resources and architectures.
- *Scalability:* Abstraction between the host system and containers gives an accurate application design, scalability and an easy-to-operate environment. Service-Oriented-Design (SoD) is combined with container applications to provide high scalability.
- *Easy dependency management:* Containers give a powerful edge to developers, enabling them to combine an application or all application components along with their dependencies as one unit. The host system doesn't face any sort of challenge regarding the dependencies required to run any application. As the host system can run Docker, everything can run on Docker containers.
- *Lightweight and isolation execution operating environments:* Though containers are not as powerful